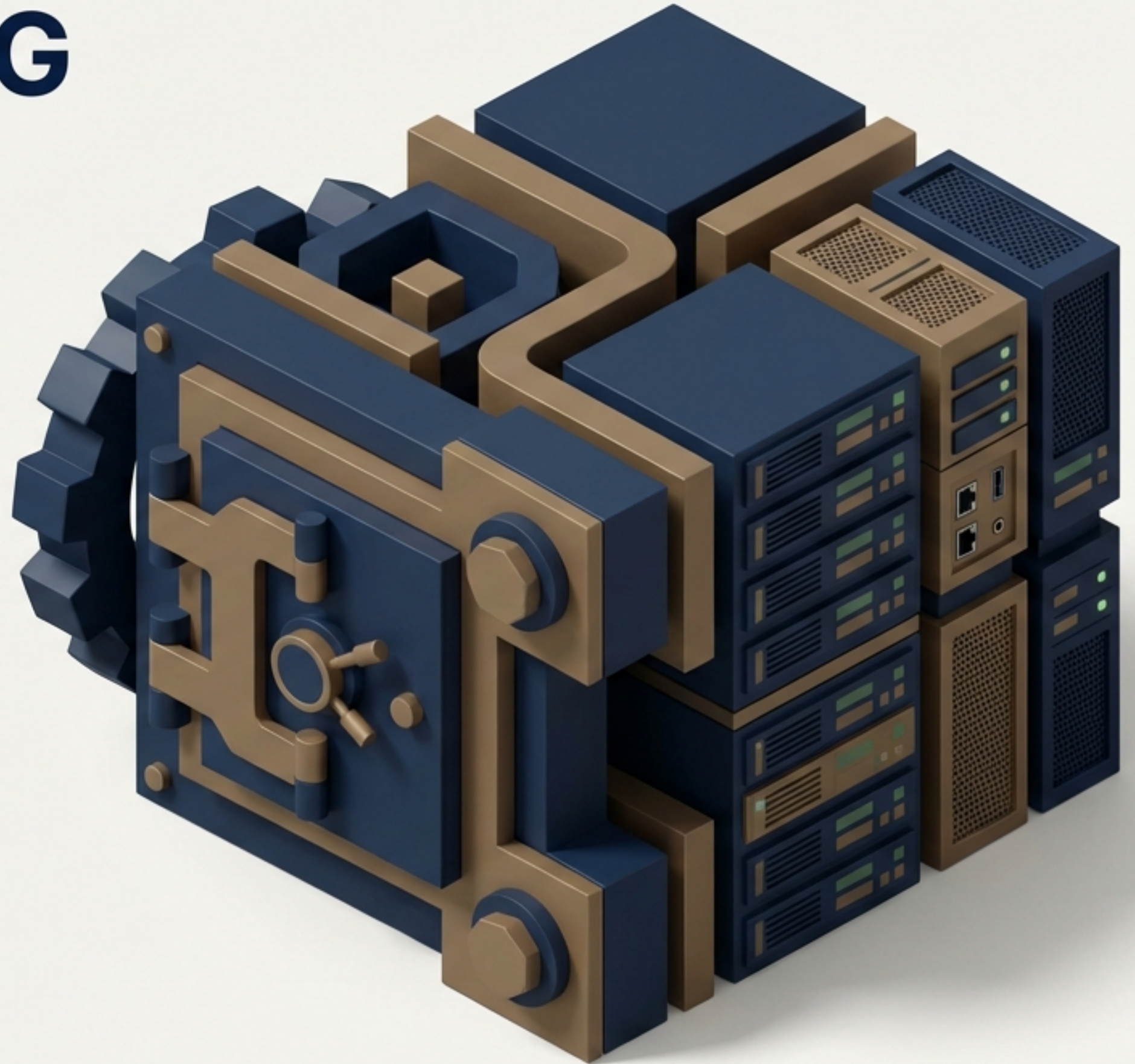


Bank-Grade RAG Architecture

**Secure, Config-Driven AI
for Enterprise Confluence**



The Bank-Grade Mandate



ACL-Aware

Strict entitlement resolution; users only query what they have permission to read.



Versioned

Content tracking ensures precise historical alignment with Confluence page updates.



Citation-Safe

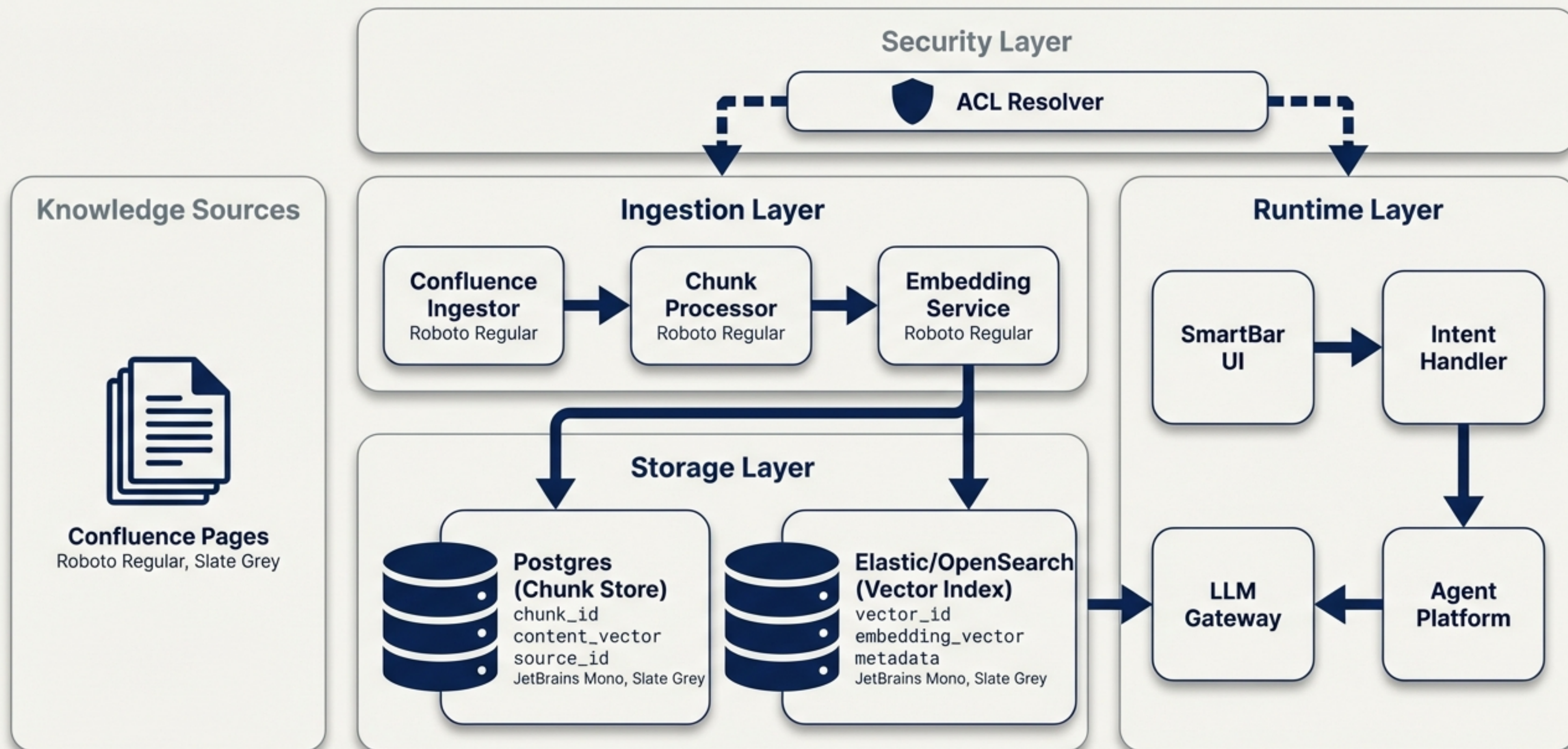
Grounded answers mapped to exact character offsets; eliminates hallucinated sources.



Auditable

Comprehensive telemetry capturing intents, chunks used, latency, and cost for compliance.

Macro Architecture Blueprint



Phase 1: The Ingestion Pipeline



The Dual-Storage Strategy



Chunk Store (Postgres)

Exact text, exact citation spans, offsets, and versioning.

```
chunkId  
pageId  
text  
char_start  
char_end
```

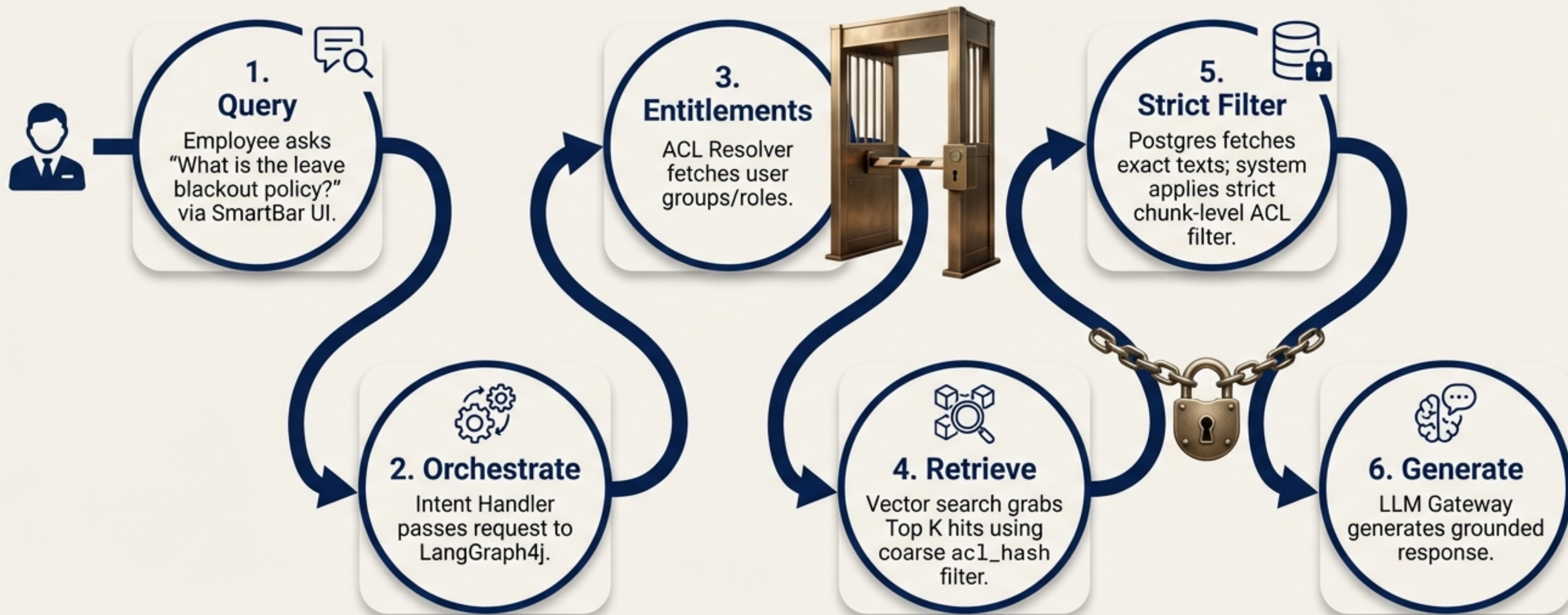


Vector Index (Elastic/OpenSearch)

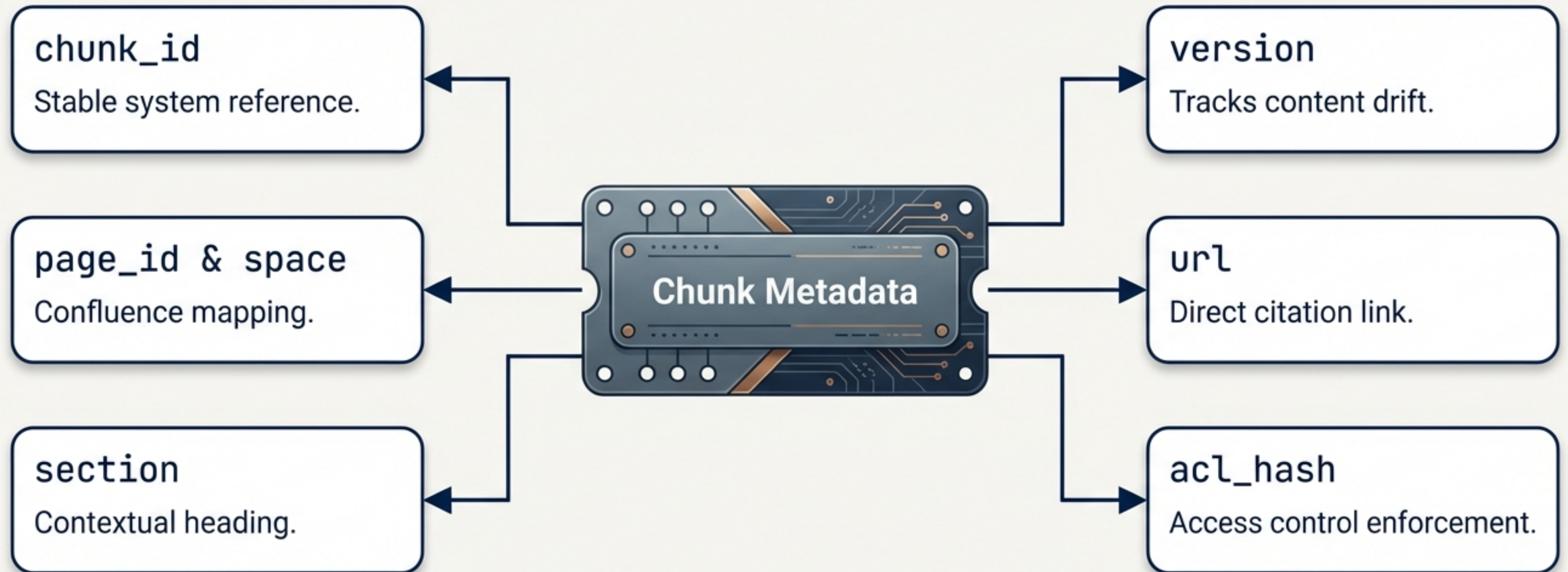
Dense vector similarity search and coarse ACL hash filtering.

```
vector[dim]  
acl_hash  
updatedAt
```


Phase 2: Runtime Query Execution



Designing Unbreakable Citations



Guarantees consistent, clickable, source-mapped citations even after database re-indexing.

The User Experience: SmartBar Source Cards

```
{  
  "intent": "kb.answer",  
  "answer": "The year-end leave  
blackout period typically applies  
from 20 December to 31 December...",  
  "citations": [  
    {  
      "citation_id": 1,  
      "snippet": "Annual leave is  
restricted...",  
      "confidence": 0.92  
    }  
  ]  
}
```

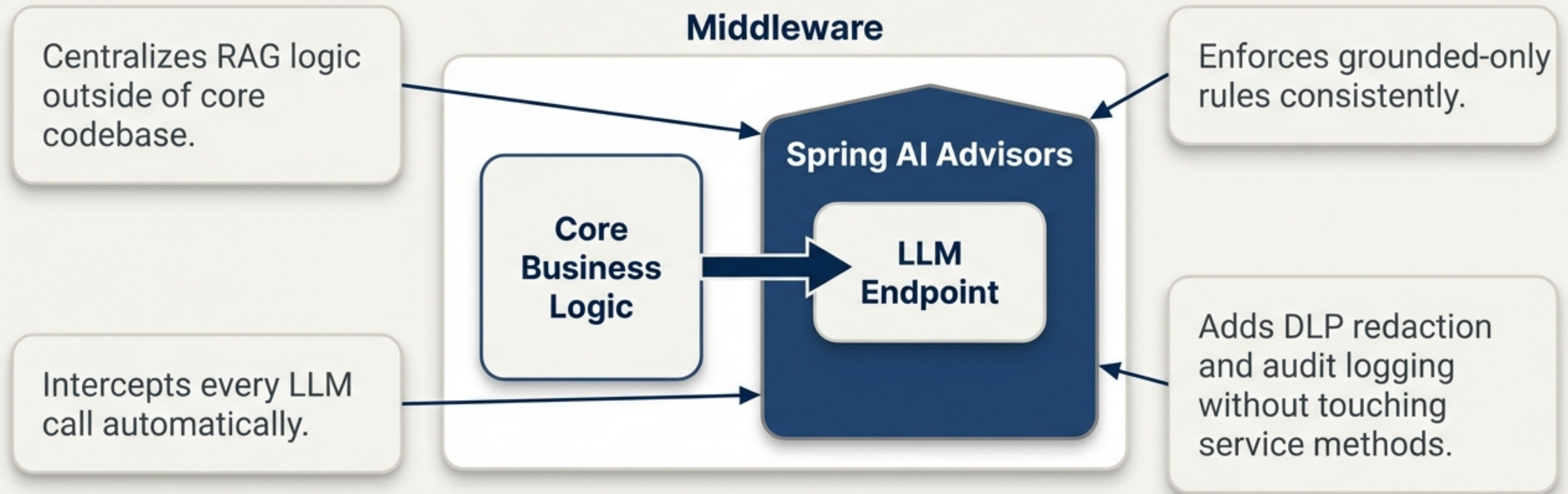
The year-end leave blackout period typically applies from 20 December to 31 December.

Employee Leave Policy

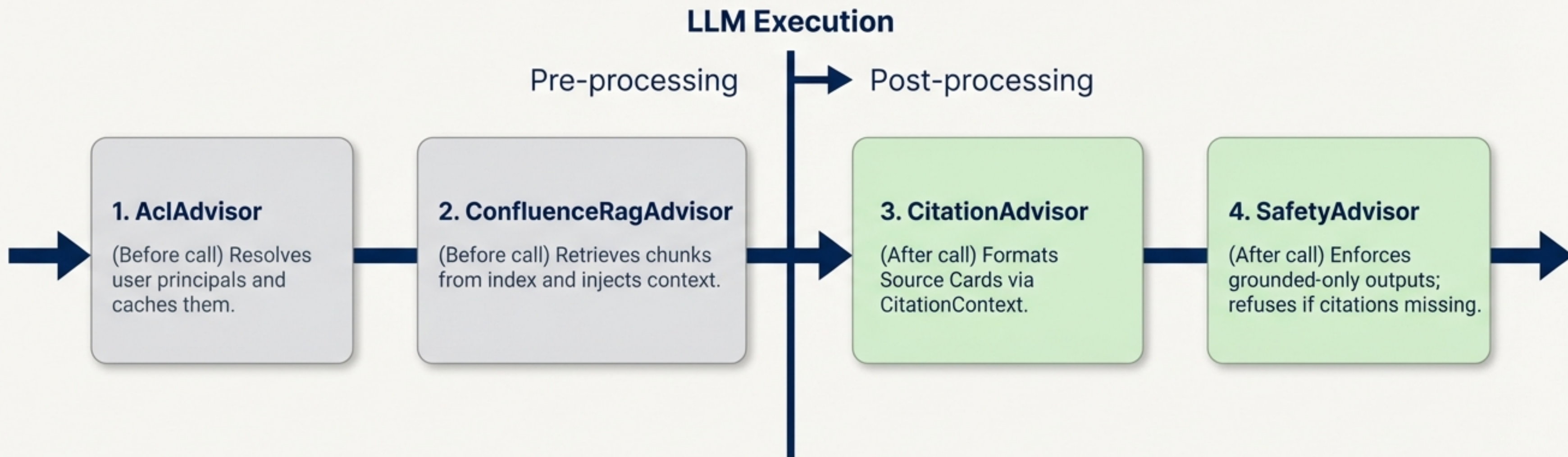
Section: Year-End Blackout Period

[View Page →](#)

The Implementation Engine: Spring AI Advisors



The Advisor Interceptor Chain



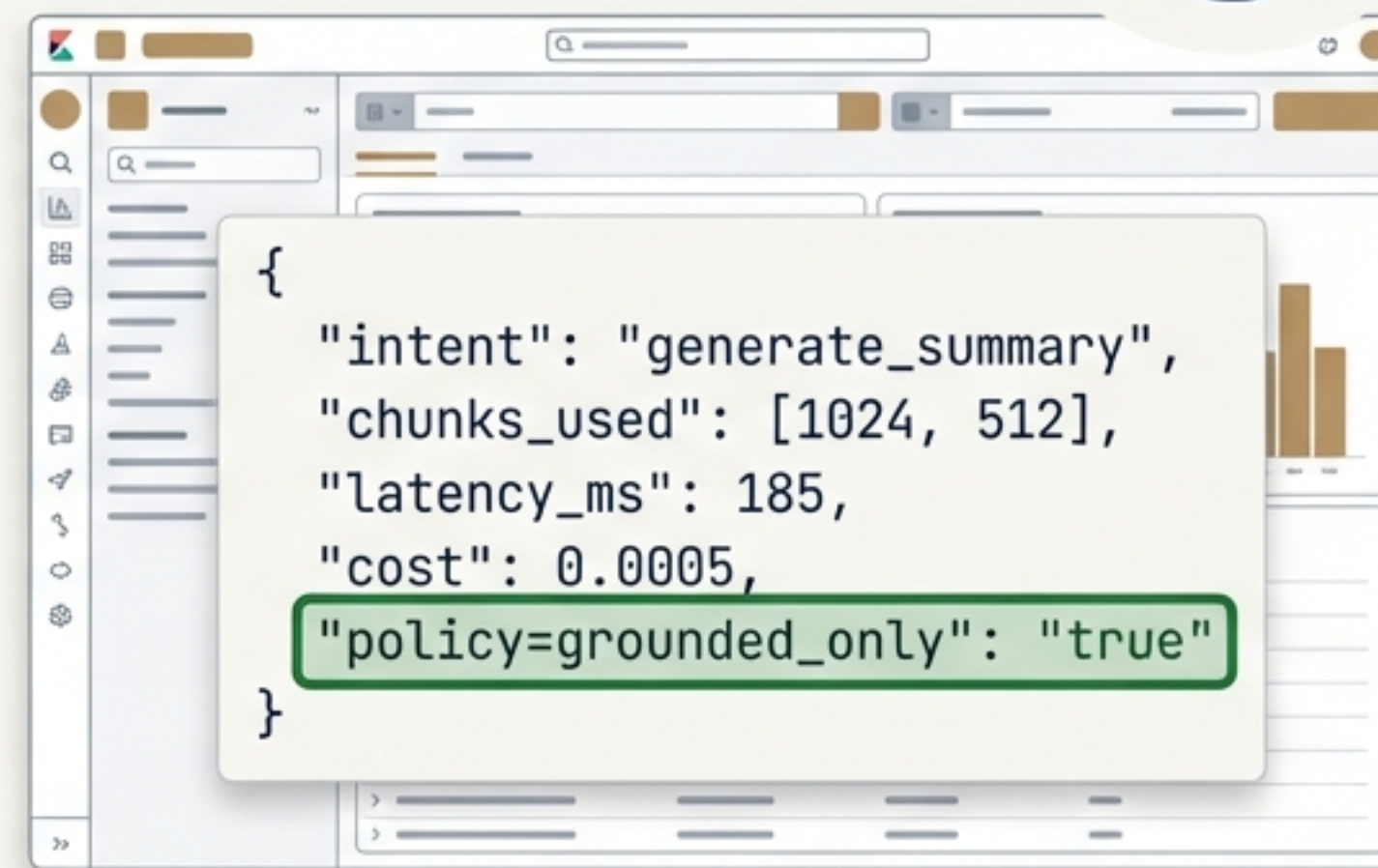
Security & Compliance Guardrails

The CITE-SAFE Prompt



1. Use **ONLY** provided context
2. If not found, say **unknown**
3. **Always** cite sources

Telemetry Logging



Architecture Summary & Execution



```
export SPRING_AI_OPENAI_API_KEY="YOUR_KEY" \  
mvn spring-boot:run
```

Config-driven Maven modules (leave_apply_v1)
enable rapid, secure deployment of enterprise-
grade AI knowledge retrieval.